INTERNAL n-GONS IN A REGULAR n-GON: THREE SIDES ARE CONCURRENT ONLY FOR $n = 3, 4, 6$, AND OTHERWISE THERE ARE $nk^2 + 1$ REGIONS

MATH0

Abstract. Divide each side of a regular n-gon into $k+1$ equal parts and, for $\mu = 1, \dots, k$, join the μ-th division points of consecutive sides; this draws k internal n-gons. We prove the two conjectures that S. R. Shannon stated in 2022 in OEIS A357216: for $n \geq 3$, some k produces a point lying on three of the nk drawn sides if and only if $n \in \{3, 4, 6\}$; and for $n = 5$ and every $n \geq 7$ the drawing cuts the polygon into exactly $nk^2 + 1$ regions, for every $k \geq 0$. A point on three sides would force a relation $\alpha W + \beta = 0$ with rational $\alpha \neq 0$, β, where $W = 1 + 2\cos(2\pi/n)$, which Niven's theorem excludes for $n = 5$ and $n \geq 7$. To our knowledge the result is new; both conjectures are still listed as open in the OEIS. The proof is checked in Lean 4 with Mathlib against a statement that math0 formalized from the OEIS entry, reviewed for fidelity and frozen before any proof attempt.

1. Introduction

Let $n \geq 3$ and $k \geq 0$. Divide each side of a regular n-gon into $k+1$ equal parts. For $\mu = 1, \dots, k$ the μ-th division points of the n sides are the vertices of a smaller regular n-gon, the μ-th *internal n-gon*. The OEIS entry A357216 [8], contributed by Scott R. Shannon on September 18, 2022, tabulates the number $T(n, k)$ of regions into which the k internal n-gons cut the outer one; A357235 [9] and A357254 [10] count vertices and edges, and the rows $n = 3, 4, 6$ are A356984, A357058 and A357196 [5, 6, 7]. Shannon's extended table covers $3 \leq n \leq 50$, $0 \leq k \leq 75$. If no three drawn segments pass through a common point, Euler's formula gives $nk^2 + 1$ regions; for $n = 3, 4, 6$ the table falls below this value (for example $T(3, 5) = 70 < 76$, $T(4, 5) = 93 < 101$, $T(6, 9) = 475 < 487$). The three entries conjecture that only the triangle, the square and the hexagon have such *non-simple intersections*, and A357216 conjectures $T(n, k) = nk^2 + 1$ for all other n. We prove both conjectures (Theorem 1); the main tool is Niven's theorem [4].

Status and prior work. A357216, A357235 and A357254 (last revised September 21, 2022) state the conjectures with no report of a proof. A356984 contains Talmon Silver's classification of the intersection points for the triangle only; A357058 and A357196 list values of k that give non-simple intersections for the square (the even ones) and the hexagon, without a general argument. The math0 referees' searches (arXiv, MathOverflow, Mathematics Stack Exchange, SeqFan, the web) found only the OEIS entries, and the conjectures are not in formal-conjectures [1]. We therefore classify the result as *new*; the argument is short and may exist somewhere unrecorded.

2. Statement

Throughout, $n \geq 3$, $k \geq 0$, indices of vertices are taken modulo n, and

$$V_j = e^{2\pi i j/n}, \qquad D_{j,\mu} = V_j + \tfrac{\mu}{k+1}\,(V_{j+1} - V_j), \qquad t_\mu = \tfrac{\mu}{k+1}.$$

Date: October 18, 2026.

The polygon is $P = \operatorname{conv}\{V_0, \dots, V_{n-1}\} \subset \mathbb{C} = \mathbb{R}^2$ and $U = \operatorname{int} P$. For $0 \le j < n$ and $1 \le \mu \le k$ the *chord* $C_{j,\mu}$ is the closed segment $[D_{j,\mu}, D_{j+1,\mu}]$: the side of the μ-th internal n-gon that cuts off the corner V_{j+1}. A *non-simple intersection* is a point lying on three distinct chords.

The entry A357216 contains, verbatim, the comment and the two formulas

```
Conjecture:  the only n-gons that contain non-simple intersections are the 3-gon
(triangle), 4-gon (square), and 6-gon (hexagon).
Conjectured formula for all columns for n >= 7:  T(n,k) = n*k^2 + 1.
Conjectured formula for all rows except for n = 3, 4, 6:  T(n,k) = n*k^2 + 1.
```

all by Scott R. Shannon, September 18, 2022.

Theorem 1. *(1) For $n \ge 3$, some $k \ge 0$ admits a point lying on three distinct chords if and only if $n \in \{3, 4, 6\}$. (2) If $n = 5$ or $n \ge 7$, then for every $k \ge 0$ the set $U \setminus \bigcup_{j,\mu} C_{j,\mu}$ has exactly $nk^2 + 1$ connected components.*

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A357216.lean`, reviewed for fidelity and frozen (SHA-256 `0b6d1419...2635391`) before any proof attempt. Its geometric definitions and the statement are, verbatim except that docstrings are omitted:

```
namespace Math0.Statements.A357216
noncomputable def vertex (n j : ℕ) : ℂ := Complex.exp (2 * Real.pi * Complex.I * j / n)
noncomputable def divPt (n k j m : ℕ) : ℂ :=
  vertex n j + ((m : ℂ) / ((k : ℂ) + 1)) * (vertex n (j + 1) - vertex n j)
noncomputable def chord (n k : ℕ) (c : Fin n × Fin k) : Set ℂ :=
  segment ℝ (divPt n k (c.1 : ℕ) ((c.2 : ℕ) + 1)) (divPt n k ((c.1 : ℕ) + 1) ((c.2 : ℕ) + 1))
noncomputable def polygon (n : ℕ) : Set ℂ :=
  convexHull ℝ (Set.range fun j : Fin n => vertex n j)
noncomputable def regionSet (n k : ℕ) : Set ℂ :=
  interior (polygon n) \ ⋃ c : Fin n × Fin k, chord n k c
noncomputable def regions (n k : ℕ) : ℕ :=
  Nat.card (ConnectedComponents (regionSet n k))
def HasNonSimpleIntersection (n k : ℕ) : Prop :=
  ∃ p : ℂ, ∃ c₁ c₂ c₃ : Fin n × Fin k, c₁ ≠ c₂ ∧ c₁ ≠ c₃ ∧ c₂ ≠ c₃ ∧
    p ∈ chord n k c₁ ∧ p ∈ chord n k c₂ ∧ p ∈ chord n k c₃
def statement : Prop :=
  (∀ n : ℕ, 3 ≤ n → ((∃ k : ℕ, HasNonSimpleIntersection n k) ↔ (n = 3 ∨ n = 4 ∨ n = 6))) ∧
  (∀ n k : ℕ, (n = 5 ∨ 7 ≤ n) → regions n k = n * k ^ 2 + 1)
```

Encoding. `vertex n j` is V_j, `divPt n k j m` is $D_{j,m}$, and the chord indexed by $c = (j, \mu - 1) \in$ `Fin n × Fin k` is $C_{j,\mu}$; since $V_n = V_0$, the chord with $j = n - 1$ ends on side 0. `regions n k` counts the components of U minus all chords, which is $T(n, k)$. `HasNonSimpleIntersection` needs no interiority condition: a point on three distinct chords lies in U, because the only points of a chord outside U are its endpoints (Lemma 5(d)) and each division point is an endpoint of only two chords (Lemma 10). The comment is read as an equivalence, which includes the existence of non-simple intersections for $n = 3, 4, 6$ that the data show; the two formulas together say $T(n, k) = nk^2 + 1$ for $n = 5$ and $n \ge 7$ and all $k \ge 0$.

Data check. The statement file also defines a computable table `T` (an exact computation in $\mathbb{Z}[2\cos(2\pi/n)]$ that uses the rotational symmetry) and its antidiagonal reading `a` with offset 3, and proves by `decide +kernel` (`a_data`, about 35 seconds of kernel time) that `a` reproduces all 66 listed terms. The identity `T n k = regions n k` is the intended meaning of `T` but is *not* proved in Lean; `statement` does not mention `T`. Outside Lean, a Python transcription of `T` and a referee's floating-point computation from the geometric definition both reproduce the entry's 195-value example table ($n \le 15$, $k \le 14$).

3. Proof

3.1. **Counting components cut out by lines.** A function $f : \mathbb{C} \to \mathbb{R}$ is *affine* if $f(z + r(w - z)) = f(z) + r(f(w) - f(z))$ for all $z, w \in \mathbb{C}$, $r \in \mathbb{R}$.

Proposition 2 (`count_general`)**.** *Let $U \subseteq \mathbb{C}$ be nonempty, open and convex, and let $f_0, \dots, f_{N-1}$ be continuous affine functions. Suppose that for each $i < N$ there are $A_i, B_i \in \mathbb{C}$ with $\gamma_i(r) = A_i + r(B_i - A_i)$ such that (i) $U \cap \{f_i = 0\} = \gamma_i((0,1))$; (ii) $f_i \not\equiv 0$; (iii) $f_j \circ \gamma_i \not\equiv 0$ for $j < i$. Let $U_N = \{z \in U : f_i(z) \neq 0 \text{ for all } i < N\}$ and $\rho_i = \#\{r \in (0,1) : f_j(\gamma_i(r)) = 0 \text{ for some } j < i\}$. Then U_N has exactly $1 + \sum_{i<N}(1 + \rho_i)$ connected components.*

Proof. Step 1 (`card_ cc_ eq`). For $z \in U_N$ let $\sigma_N(z) = \{i < N : f_i(z) > 0\}$. The set $\{y \in U : f_i(y)f_i(z) > 0 \ \forall i < N\}$ is an open neighbourhood of z in U_N on which σ_N is constant, so σ_N is locally constant, hence constant on components. Each fibre $\{\sigma_N = \sigma\}$ is convex: on a segment between two of its points each f_i is affine with the same strict sign at both ends, and U is convex. Fibres are therefore connected, and the components of U_N are exactly the nonempty fibres; their number is $|\sigma_N(U_N)|$, which is finite.

Step 2 (`ncard_ step`). Let $R_\pm = \sigma_N(\{z \in U_N : \pm f_N(z) > 0\})$. Then $\sigma_{N+1}(U_{N+1})$ is the disjoint union of $\{\sigma \cup \{N\} : \sigma \in R_+\}$ and R_-, of size $|R_+ \cup R_-| + |R_+ \cap R_-|$. If $z \in U_N$, $f_N(z) = 0$ and $f_N(w) \neq 0$ (by (ii)), then for small ε the points $z \pm \varepsilon(w - z)$ lie in the neighbourhood of Step 1, and f_N has both signs on them. Hence $R_+ \cup R_- = \sigma_N(U_N)$ and $\sigma_N(U_N \cap \{f_N = 0\}) \subseteq R_+ \cap R_-$; conversely, if $\sigma \in R_+ \cap R_-$, the zero of f_N on a segment joining points of the fibre of σ with $f_N > 0$ and $f_N < 0$ lies in that convex fibre. So $|\sigma_{N+1}(U_{N+1})| = |\sigma_N(U_N)| + |\sigma_N(U_N \cap \{f_N = 0\})|$.

Step 3 (`sgn1_ count`). By (i), $U_N \cap \{f_N = 0\} = \{\gamma_N(r) : 0 < r < 1,\ f_j(\gamma_N(r)) \neq 0 \ \forall j < N\}$. Write $f_j(\gamma_N(r)) = c_j + r d_j$; by (iii) these are nonzero affine functions of r, so the set $Z \subset (0,1)$ of their roots is finite, $|Z| = \rho_N$. By induction on $|Z|$, $s(r) = \{j < N : c_j + r d_j > 0\}$ takes exactly $|Z| + 1$ values on $(0,1) \setminus Z$: by the intermediate value theorem s is constant on root-free intervals, in particular on $(\max Z, 1)$, and this value is new, because some $c_j + r d_j$ with $d_j \neq 0$ changes sign at $\max Z$. Since $\sigma_N(\gamma_N(r)) = s(r)$, Step 2 adds $1 + \rho_N$ components, and induction on N from $|\sigma_0(U_0)| = 1$ proves the claim. □

3.2. **Corner frames.** Put $\varphi = \pi/n$. For a corner V_C, the vectors $V_{C-1} - V_C$, $V_{C+1} - V_C$ are linearly independent over $\mathbb{R}$, so every $z \in \mathbb{C}$ is uniquely $z = V_C + x(V_{C-1} - V_C) + y(V_{C+1} - V_C)$. The *frame coordinates* $x = x_C(z)$, $y = y_C(z)$ are continuous and affine; explicitly, with $w = \overline{V_C} z$, $x + y = (1 - \operatorname{Re} w)/(1 - \cos 2\varphi)$ and $y - x = \operatorname{Im} w / \sin 2\varphi$ (`fx`, `fy`, `recon`). Put

$$X(a) = \frac{\sin(a\varphi)\sin((a-1)\varphi)}{\sin\varphi\,\sin 2\varphi}, \quad Y(a) = \frac{\sin(a\varphi)\sin((a+1)\varphi)}{\sin\varphi\,\sin 2\varphi}, \quad W = 1 + 2\cos\tfrac{2\pi}{n}.$$

Lemma 3 (`fx_vertex`, `XY_range`, `Y_pos_range`)**.** *(1) For every integer a, $x_C(V_{C+a}) = X(a)$ and $y_C(V_{C+a}) = Y(a)$; X and Y have period n. (2) (X, Y) equals $(0,0)$, $(0,1)$, $(1,0)$, $(1,W)$, $(W,1)$ at $a = 0, 1, -1, 2, -2$. (3) If $a \not\equiv 0 \pmod n$ then $X(a), Y(a) \geq 0$ and $X(a) + Y(a) \geq 1$; if also $a \not\equiv -1$, then $Y(a) > 0$. In particular $W \geq 0$, and $Y(a) \geq 0$ for all a.*

Proof. (1) For $w = e^{2ia\varphi}$, double-angle formulas give $x + y = \sin^2(a\varphi)/\sin^2\varphi$ and $y - x = \sin(2a\varphi)/\sin 2\varphi$, and sum-to-product formulas give the same for $X(a) + Y(a)$ and $Y(a) - X(a)$. Replacing a by $a + n$ flips the sign of two sine factors. (2) is direct; for $a = \pm 2$ note $\sin 3\varphi / \sin\varphi = 3 - 4\sin^2\varphi = 1 + 2\cos 2\varphi$. (3) By periodicity assume $1 \leq a \leq n - 1$. Then $a\varphi \in [\varphi, \pi - \varphi]$ and $(a \pm 1)\varphi \in [0, \pi]$, so all sines are ≥ 0, $\sin(a\varphi) \geq \sin\varphi > 0$, and $X + Y = \sin^2(a\varphi)/\sin^2\varphi \geq 1$. If $a \leq n - 2$ then $(a + 1)\varphi < \pi$ and $Y(a) > 0$. Finally $W = X(-2)$. □

Lemma 4 (`polygon_sub`, `interior_pos`, `tri_interior`)**.** *For every corner C: (a) $P \subseteq \{x_C \geq 0,\ y_C \geq 0\}$; (b) $x_C(z) > 0$ and $y_C(z) > 0$ for $z \in U$; (c) $\{x_C > 0,\ y_C > 0,\ x_C + y_C < 1\} \subseteq U$. In particular U is a nonempty open convex set.*

Proof. (a) The vertices satisfy the inequalities by Lemma 3, and the right side is convex. (b) $x_C(z + e(V_C - V_{C-1})) = x_C(z) - e$, and for small $e > 0$ this point is in P, so $x_C(z) \geq e$ by (a); similarly for y_C with $V_C - V_{C+1}$. (c) Such a point is $(1 - x - y)V_C + xV_{C-1} + yV_{C+1} \in P$, and the set is open. □

3.3. **Chords and their lines.** Fix k and a chord $c = (j, \mu)$, $t = t_\mu \in (0, 1)$. By Lemma 3(2), in the frame at the corner V_{j+1} the endpoints $D_{j,\mu}$, $D_{j+1,\mu}$ have coordinates $(1-t, 0)$ and $(0, t)$, so $\gamma_c(r) = D_{j,\mu} + r(D_{j+1,\mu} - D_{j,\mu})$ has coordinates $((1-t)(1-r),\ tr)$ and lies on the zero set of the affine function

$$F_c(z) = t\, x_{j+1}(z) + (1-t)\, y_{j+1}(z) - t(1-t).$$

Lemma 5 (`gam_interior`, `line_inter`, `mem_chord_iff`)**.** *(a) $\gamma_c(r) \in U$ for $0 < r < 1$. (b) If $z \in U$ and $F_c(z) = 0$, then $z = \gamma_c(r)$ for some $r \in (0, 1)$. (c) For $z \in U$, $z \in C_c$ if and only if $F_c(z) = 0$. (d) The only points of C_c outside U are its endpoints.*

Proof. (a) Lemma 4(c): the coordinates are positive with sum $1 - r - t + 2rt < 1$. (b) By Lemma 4(b), $x, y > 0$; put $r = y/t > 0$; then $F_c(z) = 0$ gives $x = (1-t)(1-r)$, so $r < 1$, and $z = \gamma_c(r)$ because frame coordinates determine the point. (c) and (d) follow. □

For a second chord $e = (j', \mu')$ with $s = t_{\mu'}$ put $\Phi_{c,e}(r) = F_e(\gamma_c(r))$. Since $W \geq 0$, the numbers

$$a(s,t) = (1-t)\big(sW + (1-s)^2\big) + ts^2, \qquad b(s,t) = (1-t)(1-s)^2 + t\big(s^2 + (1-s)W\big)$$

are positive for $s, t \in (0, 1)$ (`aP_pos`, `bM_pos`).

Lemma 6 (`pair_same`, `pair_next`, `pair_prev`, `pair_far`)**.** *For $0 \leq r \leq 1$: (a) if $j' = j$, then $\Phi_{c,e}(r) = (t-s)(r-s)$; (b) if $j' \equiv j+1$, then $\Phi_{c,e}(r) = (1-r)\, a(s,t) + r\, s(s-t)$; (c) if $j \equiv j'+1$, then $\Phi_{c,e}(r) = (1-r)(1-s)(t-s) + r\, b(s,t)$; (d) otherwise $\Phi_{c,e}(r) > 0$.*

Proof. Let $G(a) = F_e(V_{j'+1+a}) = sX(a) + (1-s)Y(a) - s(1-s)$. Since $\gamma_c(r) = (1-r)[(1-t)V_j + tV_{j+1}] + r[(1-t)V_{j+1} + tV_{j+2}]$ and F_e is affine,

$$\Phi_{c,e}(r) = (1-r)\big[(1-t)G(\delta-1) + tG(\delta)\big] + r\big[(1-t)G(\delta) + tG(\delta+1)\big], \qquad \delta = j - j'$$

(`Fl_gam_master`). By Lemma 3(2), $G(-2) = sW + (1-s)^2$, $G(-1) = s^2$, $G(0) = -s(1-s)$, $G(1) = (1-s)^2$, $G(2) = s^2 + (1-s)W$, and substitution gives (a)–(c). In case (d), $\delta - 1, \delta, \delta + 1 \not\equiv 0$, and for $a \not\equiv 0$ Lemma 3(3) gives $G(a) \geq \min(s, 1-s)(X(a) + Y(a)) - s(1-s) \geq \min(s, 1-s) - s(1-s) > 0$. □

Corollary 7 (`cls`)**.** *If $c \neq e$, $0 < r < 1$ and $\Phi_{c,e}(r) = 0$, then one of the following holds: (S) $j' = j$ and $r = s$; (N) $j' \equiv j + 1$, $s < t$, and $(1-r)a(s,t) + rs(s-t) = 0$; (P) $j \equiv j' + 1$, $s > t$, and $(1-r)(1-s)(t-s) + rb(s,t) = 0$.*

Proof. In case (a) of Lemma 6, $t \neq s$ because $\mu \neq \mu'$. In case (b) with $s \geq t$, and in case (c) with $s \leq t$, $\Phi_{c,e}(r) > 0$. Case (d) is excluded. □

3.4. **No triple points for $n = 5$ and $n \geq 7$.**

Lemma 8 (`W_irr`, `I1`, `I2`)**.** *Let $n = 5$ or $n \geq 7$. Then $W \notin \mathbb{Q}$. Consequently, for all $t, s, \rho \in \{t_1, \ldots, t_k\}$,*

$$(1-\rho)\, a(s,t) + \rho\, s(s-t) \neq 0 \qquad \text{and} \qquad (1-\rho)(1-s)(t-s) + \rho\, b(s,t) \neq 0.$$

Proof. If W were rational, so would be $\cos(2\pi/n)$, and by Niven's theorem (Mathlib's `niven`) it would lie in $\{0, \pm\frac{1}{2}, \pm 1\}$. But $\cos(2\pi/5) \in (0, \frac{1}{2})$ because $2\pi/5 \in (\pi/3, \pi/2)$, and $\cos(2\pi/n) \in (\frac{1}{2}, 1)$ for $n \geq 7$ because $2\pi/n \in (0, \pi/3)$. The two expressions are affine in W with rational coefficients; the coefficients of W are $(1-\rho)(1-t)s \neq 0$ and $\rho t(1-s) \neq 0$. If either expression vanished, W would be rational. □

Proposition 9 (`no_triple`). *Let $n = 5$ or $n \geq 7$. No point of U is a zero of F_{c_1}, F_{c_2}, F_{c_3} for three distinct chords $c_i = (j_i, \mu_i)$.*

Proof. Suppose p is one, and write $t_i = t_{\mu_i}$. By Lemma 5(b), $p = \gamma_{c_1}(r_1) = \gamma_{c_2}(r_2)$ with $r_1, r_2 \in (0,1)$. Apply Corollary 7 to the pairs (c_1, c_2) and (c_1, c_3) at r_1. (S),(S): $j_2 = j_3 = j_1$ and $t_2 = r_1 = t_3$, so $c_2 = c_3$. (S),(N): $r_1 = t_2$ and $(1-t_2)a(t_3, t_1) + t_2 t_3(t_3 - t_1) = 0$, contradicting Lemma 8; the cases (S),(P), (N),(S) and (P),(S) are the same with the roles exchanged. (N),(N): $j_2 = j_3 \equiv j_1 + 1$. For the pair (c_2, c_3) at r_2 only (S) is possible, so $r_2 = t_3$; for (c_2, c_1) only (P) is possible (as $n \geq 3$), so $(1-t_3)(1-t_1)(t_2 - t_1) + t_3 b(t_1, t_2) = 0$, contradicting Lemma 8. (P),(P): symmetrically $j_2 = j_3$, $r_2 = t_3$, and (N) for (c_2, c_1) gives $(1-t_3)a(t_1, t_2) + t_3 t_1(t_1 - t_2) = 0$, a contradiction. (N),(P) or (P),(N): $j_3 - j_2 \equiv \pm 2$, which is not $\equiv 0, \pm 1$ as $n \geq 5$, so no case of Corollary 7 applies to (c_2, c_3) at r_2. □

3.5. **Part (1) of Theorem 1.**

Lemma 10 (`divPt_inj`). *If $D_{\sigma,\mu} = D_{\sigma',\mu'}$ with $0 \leq \sigma, \sigma' < n$ and $1 \leq \mu' \leq k$, then $\sigma = \sigma'$ and $\mu = \mu'$. Hence each division point is an endpoint of exactly two chords.*

Proof. In the frame at $V_{\sigma+1}$, $y(D_{\sigma,\mu}) = 0$ since $Y(-1) = Y(0) = 0$. If $\sigma' \neq \sigma$, let $a = \sigma' - \sigma \not\equiv 0$ and $\tau = \mu'/(k+1) \in (0,1)$; then $y(D_{\sigma',\mu'}) = (1-\tau)Y(a-1) + \tau Y(a) > 0$ by Lemma 3(3): $Y(a) > 0$ if $a \not\equiv -1$, and otherwise $a - 1 \equiv -2 \not\equiv 0, -1$ and $Y(a-1) > 0$. So $\sigma = \sigma'$, and $x(D_{\sigma,\mu}) = 1 - \mu/(k+1)$ gives $\mu = \mu'$. □

Proof of Theorem 1(1). Let $n \geq 3$, $n \notin \{3, 4, 6\}$, so $n = 5$ or $n \geq 7$, and suppose p lies on three distinct chords $c_i = (j_i, \mu_i)$ (`no_nonsimple`). If $p \in U$, then $F_{c_i}(p) = 0$ by Lemma 5(c), contradicting Proposition 9. Otherwise, by Lemma 5(d), $p = D_{\sigma_i,\mu_i}$ with $\sigma_i \in \{j_i, j_i + 1\}$ for each i. By Lemma 10 all μ_i are equal and all σ_i equal one σ; then the j_i are pairwise distinct elements of $\{\sigma, \sigma - 1\}$, which is impossible.

Conversely (`triple_exists`): if $\mu_1 \neq \mu_2$ and $(1-t_{\mu_2})a(t_{\mu_3}, t_{\mu_1}) + t_{\mu_2} t_{\mu_3}(t_{\mu_3} - t_{\mu_1}) = 0$, then $p = \gamma_{(0,\mu_1)}(t_{\mu_2}) \in U$ lies on C_{0,μ_1}, on C_{0,μ_2} by Lemma 6(a) and on C_{1,μ_3} by Lemma 6(b), using Lemma 5(c). Here $W = 0, 1, 2$ for $n = 3, 4, 6$, and the equation holds for

n	k	(μ_1, μ_2, μ_3)	$(t_{\mu_1}, t_{\mu_2}, t_{\mu_3})$	$a(t_{\mu_3}, t_{\mu_1})$	$t_{\mu_3}(t_{\mu_3} - t_{\mu_1})$
3	5	$(3, 5, 2)$	(1/2, 5/6, 1/3)	5/18	−1/18
4	5	$(4, 5, 3)$	(2/3, 5/6, 1/2)	5/12	−1/12
6	9	$(5, 9, 2)$	(1/2, 9/10, 1/5)	27/50	−3/50

as $\frac{1}{6} \cdot \frac{5}{18} = \frac{5}{6} \cdot \frac{1}{18}$, $\frac{1}{6} \cdot \frac{5}{12} = \frac{5}{6} \cdot \frac{1}{12}$, $\frac{1}{10} \cdot \frac{27}{50} = \frac{9}{10} \cdot \frac{3}{50}$ (`triple3`, `triple4`, `triple6`). □

3.6. **Part (2) of Theorem 1.** Let $n = 5$ or $n \geq 7$. Order the chords by $i = n(\mu - 1) + j$, $0 \leq i < nk$, and let $f_i = F_{(j,\mu)}$, $A_i = D_{j,\mu}$, $B_i = D_{j+1,\mu}$. By Lemma 5(c), $U \setminus \bigcup C_{j,\mu} = U_{nk}$ (`regionSet_eq`). The hypotheses of Proposition 2 hold: U by Lemma 4; (i) by Lemma 5(a),(b); (ii) since $f_i(V_{j+1}) = -t(1-t) \neq 0$; and (iii) by Lemma 6: for $e \neq c$, $\Phi_{c,e}(0)$ is $s(s-t) \neq 0$ in case (a), $a(s,t) > 0$ in case (b) and positive in case (d), and $\Phi_{c,e}(1) = b(s,t) > 0$ in case (c).

Lemma 11 (`roots_count`). *For the chord $c = (j, \mu)$ with index i, $\rho_i = 2(\mu - 1)$.*

Proof. An earlier chord $e = (j', \mu')$ has $\mu' \le \mu$. By Corollary 7, its line meets $\gamma_c((0,1))$ only in case (S), at $r = t_{\mu'}$ with $\mu' < \mu$, or in case (N), with $j' \equiv j+1$ and $\mu' < \mu$, at the unique root $r = a/(a + s(t-s)) \in (0,1)$ of the affine function in (N); case (P) needs $\mu' > \mu$. Conversely all chords (j, μ') and $(j+1, \mu')$ with $\mu' < \mu$ are earlier (their indices are $< n\mu' \le i$). These $2(\mu-1)$ roots are distinct, as a coincidence would put $\gamma_c(r)$ on three chord lines (Proposition 9). □

Proof of Theorem 1(2). By Proposition 2 and Lemma 11 (`regions_formula`, `sum_formula`), the number of components is $1 + \sum_{i<nk}\big(1 + 2\lfloor i/n \rfloor\big) = 1 + \sum_{\mu=1}^{k} n(2\mu-1) = nk^2 + 1$. □

Remark 12. For $n = 5$ and $n \ge 7$ the same lemmas give the vertex and edge counts nk^2+n and $2nk^2+n$ conjectured in A357235 and A357254: there are $nk(k-1)$ crossing points (Lemma 11), and each chord is crossed by $2(k-1)$ others at distinct points (not formalized).

4. Machine verification

The proof is in `lean/Math0/Frontier/OEIS/A357216.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. Its main theorem `Math0.Frontier.A357216.result` has type `Math0.Statements.A357216.statement`; the private auxiliary declarations are named with the results above (in Lean the chord index is $(j, \mu-1)$, and $x_C, y_C, X, Y, F_c, \gamma_c$ are `fx`, `fy`, `Xr`, `Yr`, `Fl`, `gam`). The proof does not use `T` or `a_data`. The file ends with

```
#math0_settles Math0.Frontier.A357216.result Math0.Statements.A357216.statement
```

which (see `lean/Math0/Audit.lean`) checks that `statement` is a `Prop`-valued definition without universe parameters, that `statement` and `result` use no axioms beyond `propext`, `Classical.choice` and `Quot.sound` (excluding `sorryAx`, the axioms behind `native_decide`, and custom axioms), and that the type of `result` is syntactically `statement` or its negation, with no unfolding. Running `lake env lean` on the file reports (besides linter warnings)

```
math0 audit: 'Math0.Statements.A357216.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A357216.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A357216.result' proves 'Math0.Statements.A357216.statement'
```

No `sorry`, `native_decide`, `axiom` or `set_option` occurs. Lean 4 [2] and Mathlib [3] are pinned at `v4.33.1`, and formal-conjectures [1] at commit `c77fdd4` (not used here). The script `scripts/verify.sh` rebuilds all math0 results, re-running every audit, and checks that the pinned dependencies are unmodified; for this result alone, run `lake build Math0.Frontier.OEIS.A357216` in `lean`.

References

[1] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171, 2026.

[2] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE 28, LNCS **12699**, Springer, 2021, 625–635.

[3] The mathlib Community, *The Lean mathematical library*, CPP 2020, ACM, 2020, 367–381.

[4] I. Niven, *Irrational Numbers*, Carus Math. Monographs **11**, Math. Assoc. America, 1956 (Corollary 3.12).

[5] The OEIS Foundation Inc., `https://oeis.org/A356984`, entry by S. R. Shannon (Sep. 8, 2022).

[6] The OEIS Foundation Inc., `https://oeis.org/A357058`, entry by S. R. Shannon (Sep. 10, 2022).

[7] The OEIS Foundation Inc., `https://oeis.org/A357196`, entry by S. R. Shannon (Sep. 17, 2022).

[8] The OEIS Foundation Inc., `https://oeis.org/A357216`, entry by S. R. Shannon (Sep. 18, 2022).

[9] The OEIS Foundation Inc., `https://oeis.org/A357235`, entry by S. R. Shannon (Sep. 19, 2022).

[10] The OEIS Foundation Inc., `https://oeis.org/A357254`, entry by S. R. Shannon (Sep. 20, 2022).